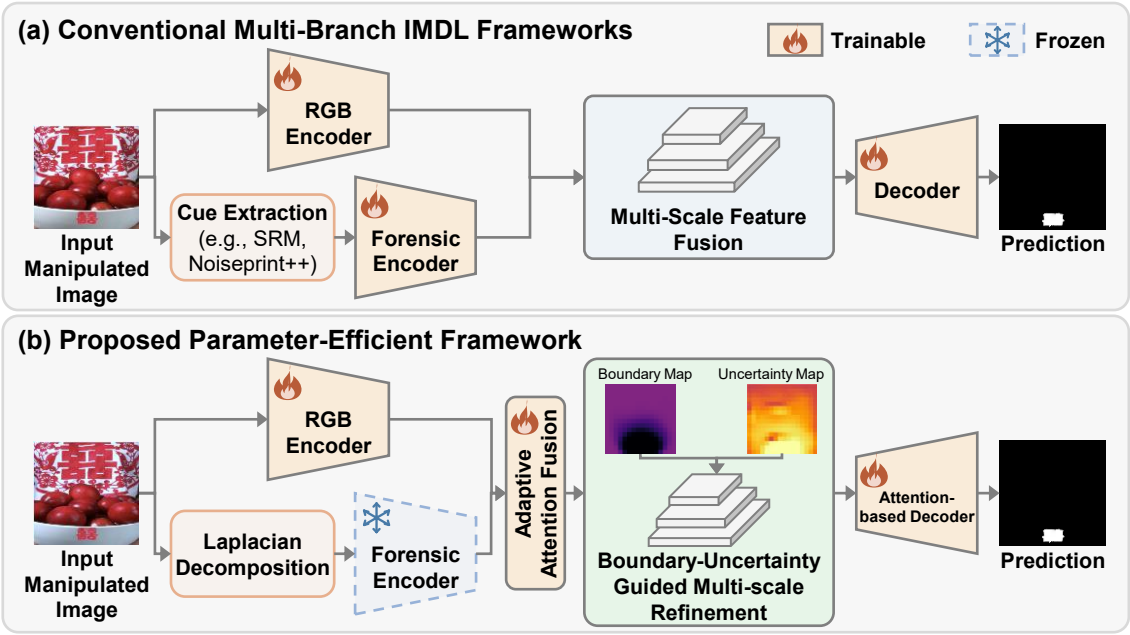

(a) Conventional Multi-Branch IMDL Frameworks
Trainable
Frozen
RGB Encoder
Cue Extraction (e.g., SRM, Noiseprint++)
Forensic Encoder
Input Manipulated Image
Multi-Scale Feature Fusion
Decoder
Prediction
(b) Proposed Parameter-Efficient Framework
RGB Encoder
Laplacian Decomposition
Forensic Encoder
Adaptive Attention Fusion
Boundary Map
Uncertainty Map
Boundary-Uncertainty Guided Multi-scale Refinement
Attention-based Decoder
Input Manipulated Image
Prediction